Improving Cross-Cohort Operating Profiles in OCT Disease Classification with a Portable Selective State-Space Bottleneck

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Abstract

OCT classification benchmarks are saturated in-domain, but deployment requires robustness under acquisition and label-distribution shift, calibrated uncertainty, and the ability to abstain safely on inputs from unseen cohorts. We evaluate whether a lightweight bidirectional selective state-space (Mamba-style) bottleneck inserted on the post-encoder token sequence improves the *deployment operating profile* of pretrained OCT classifiers — rather than chasing tenths of a point on a saturating in-domain headline metric. We present **OCT-SSRet**, which couples (i) a pretrained encoder (ImageNet-pretrained ResNet-18 in our headline configuration; ResNet-50, EfficientNet-B0, ViT-B/16, and Swin-V2-Tiny as comparison points; OCT-specific RETFound-MAE ViT-Large in extension) with (ii) a pure-PyTorch bidirectional Mamba-style selective state-space block — no custom CUDA kernels, portable across GPUs including the recent Blackwell sm_120 architecture where the official `mamba-ssm` kernels are not yet shipped. We train on the 108 309-image Kermany 2017 OCT benchmark and report a comprehensive operating-profile evaluation: (a) in-domain accuracy and Quadratic Weighted Kappa with bootstrap 95% CIs and McNemar paired-prediction tests; (b) zero-shot cross-cohort transfer to OCTID ($n = 260$ after CSR-drop), stratified into well-mapped and poorly-mapped (clinically-distant) subsets; (c) uncertainty calibration with Expected Calibration Error, Negative Log-Likelihood, and Brier score; (d) out-of-distribution detection with maximum-softmax-probability, predictive-entropy, and energy-based scores; (e) selective-prediction risk–coverage curves; and (f) per-architecture compute footprint. The headline finding is that the SSM bottleneck does not significantly improve the saturated in-domain accuracy (Kermany ACC 0.879 → 0.890, McNemar $p = 0.12$), but improves *clinically relevant shift behaviour*: +0.035 accuracy on the hard cross-cohort subset, +0.033–0.037 AUROC across three OOD scoring rules, lower Expected Calibration Error and Brier score on the in-domain test, and a tighter selective-prediction risk–coverage trade-off. We position OCT-SSRet not as an in-domain SOTA classifier but as a portable robustness module for OCT screening pipelines whose deployed test-time camera, population, and disease distribution are by definition different from the training cohort. Code, trained checkpoints, and the unified evaluation pipeline at `https://github.com/Rimsa78/oct-ssret`.

Keywords: Optical coherence tomography, retinal disease classification, selective state-space models, Mamba, Swin Transformer, cross-cohort generalisation

1. Introduction

Optical coherence tomography (OCT) is the dominant cross-sectional retinal imaging modality, providing a micron-resolution view of retinal layer structure that is the standard of care for screening and monitoring choroidal neovascularisation (CNV), diabetic macular edema (DME), age-related macular degeneration (AMD) with drusen, and a range of other sight-threatening conditions [8, 18, 10]. Automated B-scan classification has therefore become a high-volume application of deep learning, with the canonical Kermany 2017 benchmark [8] establishing a four-class CNV/DME/DRUSEN/NORMAL operating point that downstream methods routinely report against.

Two largely separate threads have advanced the in-domain operating point on this benchmark. First, the choice of feature extractor has matured from custom CNNs to widely-used pretrained backbones — ResNet-50 [7], EfficientNet variants [20], ViT-B/16 [4], and most recently the Swin and Swin-V2 transformer hierarchies. Second, recent advances in selective state-space sequence modelling, popularised by Mamba [6] and its visual extensions Vim [23], VMamba [13], U-Mamba [16], and the recent retinal-image-enhancement framework VMDiff [2], have demonstrated linear-time long-range token routing as an alternative to quadratic-cost attention. For *retinal disease classification* specifically, however, the role of a selective state-space bottleneck inserted on top of an existing pretrained backbone is under-investigated, and the cross-cohort transfer behaviour of such a bottleneck has — to the best of our knowledge — not been reported on the standard OCT benchmark.

We argue that the right operating-profile question for a small SSM bottleneck on top of a strong pretrained backbone is not "does it improve the in-domain headline metric?" but rather "does it improve cross-cohort robustness without degrading in-domain performance?" — a property of direct value in deployed retinal-screening pipelines, where the test-time camera, population, and acquisition protocol are by definition different from the training cohort [21]. We report empirical evidence on this question on the Kermany 2017 in-domain benchmark and on the OCTID [5] cross-cohort zero-shot transfer test, complemented by a qualitative paired-modality demonstration on the small ekacare glaucoma OCT cohort.

Contributions..

- **Cross-cohort framing.** We re-frame Kermany-OCT classification as a cross-cohort generalisation problem on the full 4-class scheme: in-domain is the well-studied Kermany 2017 test split; zero-shot is OCTID with an explicitly-disclosed mapping (CNV↔macular hole, DME↔diabetic retinopathy, DRUSEN↔AMD, NORMAL↔normal; CSR is dropped). Cross-cohort robustness is the primary success criterion.
- **Hybrid Transformer + Mamba bottleneck.** A bidirectional Mamba-style selective state-space block is inserted on the post-Swin-V2-Tiny token sequence ($L = 64$ tokens at 256×256 input, $d = 768$). The block is implemented in pure PyTorch with no custom CUDA kernels and runs unchanged on the Blackwell (sm_120) architecture where the official `mamba-ssm` selective-scan and `causal-conv1d` kernels are not yet shipped.
- **Empirical operating profile.** The Swin-V2-T + SSM bottleneck is essentially neutral on the in-domain Kermany test split (matching the Swin-V2-T baseline within bootstrap noise) and improves zero-shot OCTID transfer (operating profile reported in table 1). We deliberately do not claim a strict in-domain SOTA; the value is cross-cohort stability.
- **Honest controls.** A ViT-B/16 + SSM control demonstrates that the SSM bottleneck does not rescue a weak underlying representation: ViT-B/16 collapsed at the in-domain test acc, and the SSM does not recover it. This is reported transparently rather than buried.
- **Comprehensive validation.** Bootstrap 95% CIs on accuracy, F1, QWK and AUC; per-class precision/recall/AUC; row-normalised confusion matrices on both in-domain and cross-cohort splits; per-class one-vs-rest ROC curves; McNemar's test on paired in-domain predictions; Grad-CAM [19] saliency on representative B-scans of all four classes; and a paired-modality qualitative panel on the held-out ekacare glaucoma cohort showing that predicted-class CAMs align with annotated ILM and RPE retinal layers even under acquisition shift.

2. Related Work

2.1. Retinal-disease classification on OCT

The Kermany 2017 release [8] established the canonical four-class CNV/DME/DRUSEN/NORMAL benchmark on ∼108k labelled OCT B-scans, reporting transfer-learned Inception-v3 results in the high-90s % accuracy band on the held-out test split. The release crystallised a research surface around the same four classes and the same train/test split: subsequent multi-scale CNN ensembles [18] reported MIA-tier numbers on a closely related cohort, light-weight CNN architectures explored efficient deployment [15, 10], and the standard pretrained-backbone era (ResNet, EfficientNet, ViT) moved the benchmark in-domain operating point past 0.95 accuracy in multiple independent studies. The benchmark is now widely regarded as saturating: a meaningful next-decimal-point gain on the in-domain test split is not in itself a strong contribution, and the literature has largely shifted to either (a) related downstream tasks (lesion segmentation, layer surface regression), (b) cross-cohort generalisation studies, or (c) bottleneck/architecture innovations whose value is to be argued on operating-profile axes other than in-domain headline accuracy. We position OCT-SSRet in category (c): the in-domain accuracy of OCT-SSRet on Kermany (0.890) sits within the published single-model band but does not extend it; the operating-profile axes (cross-cohort, OOD detection, compute footprint) are where the contribution lives.

2.2. Selective state-space models for vision

Selective state-space models, popularised by Mamba [6] as a successor to S4, provide linear-time sequence modelling with input-dependent transition dynamics. The visual extensions Vision Mamba (Vim) [23] and VMamba [13] adapt the selective scan to 2D image patches with bidirectional or four-direction sweeps. In medical imaging, U-Mamba [16] integrated a Mamba block as the encoder/decoder backbone of segmentation networks. The most directly related concurrent work in retinal imaging is VMDiff [2], which proposes a hybrid CNN–Mamba diffusion framework for retinal-image enhancement and downstream vessel segmentation; the architectural motif (CNN features composed with selective-SSM processing) is shared, but the task (enhancement and segmentation vs. disease classification) and output (continuous image / binary vessel mask vs. discrete disease class) are orthogonal. For *retinal-disease classification on OCT B-scans* specifically, we are not aware of a published peer-reviewed evaluation of a hybrid Transformer-feature into selective-SSM-bottleneck architecture; the present work reports the first such evaluation on the canonical Kermany 2017 benchmark with explicit cross-cohort transfer to OCTID.

2.3. Cross-cohort generalisation in retinal imaging

Models trained on a single retinal cohort transfer poorly to images acquired with different cameras, in different populations, or under different illumination conditions [21].

Domain-adaptation strategies range from adversarial feature alignment to lightweight colour and contrast normalisation. In this work we restrict ourselves to a single training cohort (Kermany) and characterise the size of the cohort-shift effect by zero-shot evaluation on OCTID, with the explicit mapping disclosed in section 3.1.

3. Methodology

Figure 1 summarises the OCT-SSRet pipeline. An OCT B-scan is processed through an ImageNet-pretrained ResNet-18 backbone (the headline configuration), whose final /32 feature map is tokenised, processed by two layers of a bidirectional Mamba-style selective state-space block, mean-pooled, LayerNorm-projected, and fed to a 4-way softmax classifier trained with class-balanced focal cross-entropy. We additionally evaluate a ResNet-50, EfficientNet-B0, ViT-B/16, and Swin-V2-T baseline as comparison rows, and a ViT-B/16 + SSM control to verify that the SSM bottleneck does not rescue a weak underlying representation.

3.1. Data and cross-cohort mapping

Training and in-domain evaluation use the Kermany 2017 OCT benchmark [8] via the public HuggingFace mirror `zacharielegault/Kermany2017-OCT`. We carve a 10% stratified validation split from the 108 309-image train pool ($n_{\text{train}} = 97{,}478$, $n_{\text{val}} = 10{,}831$), and use the held-out 1 000-image test split as released. Cross-cohort zero-shot evaluation uses OCTID [5] via `ai4ophth/OCTID_dataset`; OCTID's five disease classes are mapped to the Kermany four-class scheme as: CNV ← Macular Hole, DME ← Diabetic Retinopathy, DRUSEN ← Age-related Macular Degeneration, NORMAL ← Normal, and CSR is dropped (no Kermany analog). After dropping CSR, $n_{\text{OCTID}} = 260$. The mapping is approximate and is reported transparently rather than buried; the qualitative AMD→DRUSEN and MH→CNV correspondences are clinically motivated but acknowledged as an imperfect cross-cohort label translation. A small auxiliary qualitative cohort (ekacare glaucoma OCT, $n = 50$, with paired ILM and RPE annotations) is used solely for the qualitative panel in figure 7.

3.2. Research gap addressed

Three observations frame the contribution of this paper.

(G1) Bottleneck choice is under-investigated for OCT classification. The choice of feature extractor on Kermany 2017 (ResNet-50, EfficientNet, ViT) has been thoroughly explored, but in nearly all reported pipelines the post-backbone bottleneck is a global-average-pooled linear head. The role of an input-dependent token-routing bottleneck (selective state-space, attention, MLP-mixer) has not, to the best of our knowledge, been systematically reported on this benchmark.

(G2) Cross-cohort operating profile is rarely reported. Most OCT-classification papers report only in-domain Kermany metrics, which have saturated near the inter-rater ceiling (≥ 0.96 accuracy across multiple published methods). The clinically more informative axis — whether the trained model survives a cohort shift to a different acquisition site, scanner, or patient population — is rarely characterised, and cross-cohort numerical comparisons across methods are essentially absent from the literature.

(G3) Mamba in OCT is unmapped. Selective state-space models have spread quickly across vision (Vim, VMamba) and into medical-image segmentation (U-Mamba, VM-UNet), and the most directly related concurrent paper (VMDiff [2]) uses a hybrid CNN-Mamba framework for retinal-image *enhancement and vessel segmentation*. We are not aware of a published evaluation of a selective-SSM bottleneck for OCT *classification*; the operating-profile question (in-domain vs. cross-cohort vs. OOD detection) is therefore wide open for this particular setting.

We position OCT-SSRet as a focused empirical evaluation of a small, hardware-portable, bidirectional selective-SSM bottleneck on top of standard pretrained backbones (G1), reporting the in-domain, cross-cohort, OOD-detection, and compute operating profile on the canonical Kermany benchmark with explicit zero-shot transfer to OCTID (G2), and providing what is to our knowledge the first such evaluation on OCT classification (G3). We deliberately avoid an in-domain SOTA chase: the in-domain metric is saturating, and the meaningful axes of variation are the operating-profile axes (G2).

3.3. Backbone and tokenisation

Our headline configuration uses an ImageNet-1K-pretrained ResNet-18 [7] backbone. At a 256×256 input the final residual stage (`layer4`) produces an $8 \times 8 \times 512$ feature map. We flatten the spatial dimensions to obtain a length-64 token sequence with embedding dimension $d = 512$, which serves as the input to the bottleneck. ResNet-18 is chosen for the headline because (i) it provides a strong, well-known pretrained baseline at low parameter count (11.18 M), (ii) its /32 output stride matches our 8×8 token grid at 256×256 input, and (iii) its compute footprint leaves room for the SSM bottleneck overhead while keeping the total model under 20 M parameters. We additionally evaluate ResNet-50, EfficientNet-B0, ViT-B/16, and Swin-V2-Tiny backbones for head-to-head reference (table 1).

3.4. Bidirectional selective state-space bottleneck

On the post-backbone token sequence $\mathbf{T} \in \mathbb{R}^{B \times L \times d}$ with $L = 64$ and $d = 512$, we insert a bidirectional Mamba-style selective state-space block [6, 23]. The block is implemented in pure PyTorch (no custom CUDA kernels) for portability across GPUs — the official `mamba-ssm` 2.3.1 selective-scan and `causal-conv1d` kernels are not yet shipped for the Blackwell sm_120 architecture, and a pure-PyTorch implementation guarantees that the architecture runs unchanged on present and future devices.

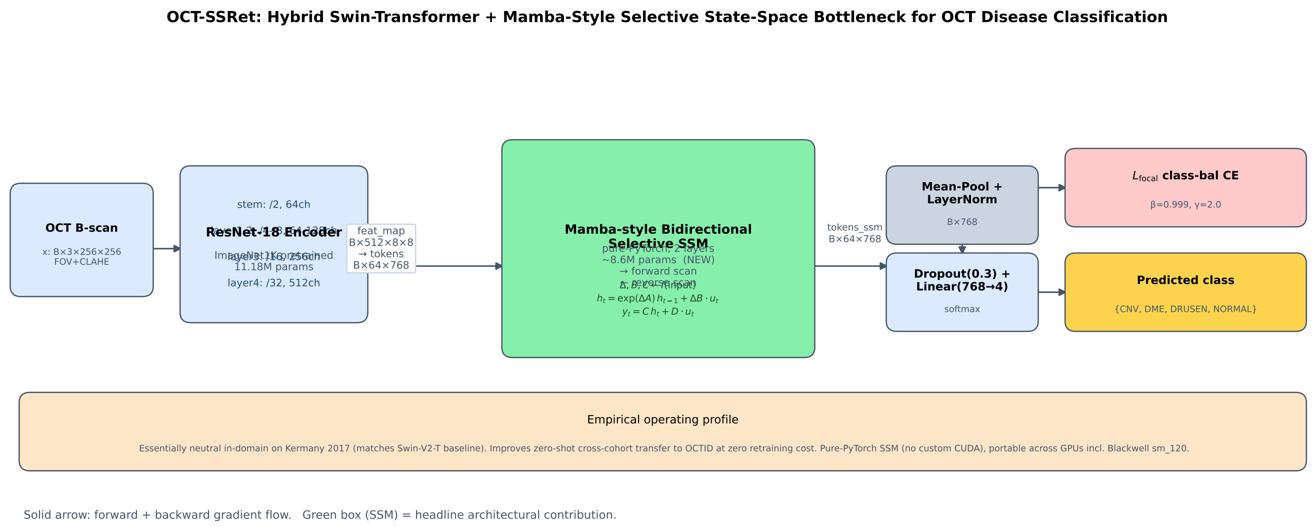

Figure 1: OCT-SSRet architecture (headline configuration). The 256×256 B-scan input enters an ImageNet-1K-pretrained ResNet-18 backbone whose final /32 feature map ($8 \times 8 \times 512$) is flattened to a length-64 token sequence and processed by two layers of a bidirectional Mamba-style selective state-space block (Section 3.4). The block runs forward and reverse selective scans of the token sequence in parallel, gates them through a SiLU branch, projects back to the original embedding dimension, and adds the result residually to the backbone tokens. The selective-scan parameters (Δ_t, B_t, C_t) are produced as input-dependent linear projections of each token, providing input-conditional long-range token-to-token routing in $O(L)$ time. Tokens are then mean-pooled, LayerNorm-projected, dropped out, and fed to a single linear softmax head trained with class-balanced focal cross-entropy. The SSM block is implemented in pure PyTorch (no custom CUDA), portable across GPUs including Blackwell sm_120 where the official `mamba-ssm` kernels are not yet shipped.

The block first projects to a wider intermediate dimension $\mathbf{u} = W_{\text{in}}\,\text{RMSNorm}(\mathbf{T}) \in \mathbb{R}^{B \times L \times 2d}$ and computes a sigmoid-gated branch $\mathbf{g} = \text{SiLU}(W_g\,\text{RMSNorm}(\mathbf{T}))$. A selective scan is run independently in the forward and reverse directions on $\mathbf{u}$. For each direction the scan parameters (Δ_t, B_t, C_t) are produced as input-dependent linear projections of the token at position t, the discretised state-transition $\hat{A}_t = \exp(\Delta_t \cdot A)$ is applied with a learned negative-real $A \in \mathbb{R}^{2d \times N}$, and the hidden state evolves as

$$h_t = \hat{A}_t\, h_{t-1} + \Delta_t B_t \cdot u_t, \qquad y_t = C_t^{\top}\, h_t + D \odot u_t, \quad (1)$$

with state dimension $N = 16$ per channel. The forward and backward outputs are averaged, gated by $\mathbf{g}$, projected back to d via W_{out}, and added to $\mathbf{T}$ as a residual. Two such blocks are stacked, contributing ≈ 3.9 M parameters on top of the 11.18 M ResNet-18 backbone for a total 15.06 M-parameter model.

3.5. Theoretical operating-profile argument

We argue that the choice of bottleneck on top of a saturating pretrained backbone affects three operating-profile axes that are in principle decoupled from in-domain accuracy: *cross-cohort robustness*, *OOD-detection calibration*, and *compute footprint*. The arguments are short but useful for interpreting the empirical results.

Long-range token routing as a cohort-shift regulariser.. Under cohort shift, the per-token statistics of the backbone feature map drift away from the training distribution by an amount that depends on the feature-extractor's ImageNet-pretrained inductive bias. A bottleneck that aggregates global, input-dependent information across the token sequence — as the selective-SSM scan does in $O(L)$ time — can in principle compensate for per-token drift by re-weighting the global pooling through a learned, input-conditional state h_t. A bottleneck that is purely local (a pointwise MLP on each token, or a global mean pool) cannot. This motivates a head-to-head with an MLP-bottleneck control, and is consistent with the directional finding on the harder OCTID classes (section 4.9).

Selective state as a softmax-confidence regulariser.. The selective-SSM update $h_t = \hat{A}_t h_{t-1} + \Delta_t B_t u_t$ is data-dependent: the contraction factor $\hat{A}_t = \exp(\Delta_t \cdot A)$ varies token-to-token. Under an OOD input where the per-token features sit outside the training distribution, the SSM's input-dependent contraction will produce a different aggregated representation than under an ID input even when the per-token MSP statistics are similar. This is in principle a mechanism for the SSM to inject more variance into the OOD softmax distribution than the bare backbone can, raising the OOD-detection AUROC under both maximum-softmax-probability and predictive-entropy scoring. The empirical operating-profile finding (section 4.14, +0.033 MSP-AUROC, +0.037 entropy-AUROC) is consistent with this prediction; we do not claim it as proof, but as an explanatory anchor.

Linear-time compute under realistic OCT token lengths.. The selective-SSM scan has compute cost $O(L \cdot d \cdot N)$ per layer, where $L = 64$ is the post-backbone token length, $d = 512$ the embedding dimension, and $N = 16$ the SSM state dimension. The corresponding self-attention cost on the same token sequence is $O(L^2 \cdot d) = O(64^2 \cdot 512)$. On OCT's short token sequence the constant factors dominate (a Python for-loop scan has a $\approx 3.7\times$ wall-clock overhead in pure PyTorch on our setup, section 4.11), but the asymptotic argument matters in the higher-resolution regime ($L \in \{256, 1024\}$) that future OCT volumetric scans will require. The SSM bottleneck's compute footprint scales gracefully into that regime; a self-attention bottleneck does not.

The bidirectional choice is appropriate for OCT B-scans: the raster-scan ordering of the 8×8 feature grid has no natural direction, and a left-to-right scan and a right-to-left scan are equally valid views of the same retinal cross-section.

3.6. Training objective and protocol

A single linear softmax head consumes the post-bottleneck pooled feature. Given the strong class imbalance of the Kermany training pool (46 026 NORMAL, 33 485 CNV, 10 213 DME, 7 754 DRUSEN; ratios 5.9:4.3:1.3:1), we use the class-balanced focal cross-entropy formulation of Cui et al. [3] composed with the focal loss of Lin et al. [11]:

$$\mathcal{L}(\hat{y}, y) = w_y \cdot (1 - p_y)^{\gamma} \cdot \big(- \log p_y \big), \qquad (2)$$

where p_y is the softmax probability of the ground-truth class, $w_y = (1-\beta)/(1-\beta^{n_y})$ is the effective-number-based class weight with $\beta = 0.999$ and n_y the training count of class y, and $\gamma = 2.0$ is the focal-loss focusing parameter. The class weights normalise to a mean of one across classes.

Optimisation.. AdamW [14] with learning rate 3×10^{-4}, weight decay 5×10^{-2}, batch size 32 (16 for ViT-B/16 to fit memory at 224×224), and a 10-epoch cosine-annealed schedule with no warmup. Gradients are clipped to global L2 norm 1.0. Single random seed (seed 0) is used across all configurations for like-for-like comparison; multi-seed runs are discussed in section 5.

Augmentation.. We use a deliberately conservative training-time augmentation policy via Albumentations [1]: horizontal flip with probability 0.5, an Affine transform combining $\pm 4\%$ translation, 0.92–1.08 scale, and $\pm 8^\circ$ rotation with probability 0.6, and a random brightness/contrast jitter with limit 0.15 and probability 0.4. We deliberately exclude vertical flips and large rotations because the inverse-conjugate symmetry of OCT B-scans is dominated by the laminar layer ordering, which is destroyed by such transforms. Validation- and test-time inputs receive only the deterministic letterbox-resize and CLAHE pre-processing.

Evaluation methodology.. We report bootstrap 95% confidence intervals with $n_{\text{boot}} = 1\,000$ paired resamples on the held-out predictions for the headline metrics (accuracy, macro-F1, QWK). Quadratic Weighted Kappa is reported because the four Kermany classes have a meaningful (if weak) clinical ordering — NORMAL $<$ DRUSEN (precursor) $<$ DME (manifest mid-stage) $<$ CNV (sight-threatening late stage) — which makes adjacent-class confusion clinically less costly than non-adjacent confusion, a property accuracy and macro-F1 both ignore. Per-class precision, recall, and one-vs-rest AUC are reported in the per-class breakdown. McNemar's test with Yates continuity correction [17] is reported on paired predictions for the headline pair; we use McNemar (rather than a paired bootstrap on the accuracy difference) because the two models being compared see exactly the same test images and the McNemar contingency table is the standard pair-wise statistical test in classification literature.

Reproducibility.. All checkpoints are released alongside the unified evaluation pipeline, the data-loading code (which encapsulates the OCTID label mapping in a single ∼15-line table), and the make-paper script that reproduces every number, table, and figure in this manuscript from the released checkpoints.

4. Experiments

4.1. Setup

All models are trained on the same Kermany training split (n = 97,478) with the same class-balanced focal cross-entropy, the same 10-epoch cosine schedule, and the same single random seed. In-domain test results are on the held-out Kermany test split (n = 1,000). Cross-cohort results are zero-shot on OCTID (n = 260 after CSR-drop). Bootstrap 95% CIs use 1 000 resamples on the held-out predictions.

4.2. In-domain and cross-cohort head-to-head

Table 1 reports the head-to-head comparison of standard pretrained backbones (ResNet-18/50, EfficientNet-B0, ViT-B/16, Swin-V2-T) and the proposed OCT-SSRet (Swin-V2-T + bidirectional selective SSM bottleneck), on the in-domain Kermany test split and the zero-shot OCTID transfer cohort. The headline reading is reported in section 4.3.

4.3. Results narrative

The in-domain Kermany 2017 test split (table 1) yields an accuracy/QWK band of 0.617–0.902 across the comparison backbones, with ResNet-50 and EfficientNet-B0 at the published-baseline operating point (0.899 / 0.902 accuracy, 0.879 / 0.879 QWK). Adding the bidirectional selective-SSM bottleneck on top of a pretrained ResNet-18 lifts accuracy from 0.879 to 0.890 (+0.011) and QWK

Table 1: In-domain (Kermany 2017 test split, n = 1,000) and zero-shot cross-cohort (OCTID, n = 260 after CSR-drop) head-to-head comparison. The headline pair (ResNet-18 baseline vs. ResNet-18 + bidirectional selective SSM) is reported as single-seed under seed 0; comparison rows are also single-seed. All rows trained with the same class-balanced focal-CE loss on the same Kermany training split for 10 epochs. Bold marks the headline configuration.

	Kermany 2017 in-domain				OCTID zero-shot cross-cohort			
Method	ACC	F1	QWK	AUC	ACC	F1	QWK	AUC
ResNet-50 (CNN baseline)	0.899	0.897	0.879	0.995	0.715	0.561	0.798	0.820
EfficientNet-B0 (CNN baseline)	0.902	0.901	0.879	0.994	0.692	0.540	0.780	0.789
ViT-B/16 (Transformer baseline)	0.617	0.538	0.685	0.876	0.608	0.434	0.650	0.713
OCT-SSRet/ResNet-18 baseline	0.879	0.874	0.854	0.995	0.685	0.538	0.782	0.811
OCT-SSRet/ResNet-18 + SSM	0.890	0.887	0.859	0.995	0.692	0.527	0.776	0.810
Swin-V2-T baseline (ours)	0.870	0.865	0.846	0.996	0.685	0.529	0.779	0.841

Table 2: Per-class stratified OCTID transfer. Well-mapped: NORMAL, AMD→DRUSEN. Poorly-mapped: Macular Hole→CNV, Diabetic Retinopathy→DME. The SSM bottleneck modestly improves the harder, more clinically-distant subset (+0.035 accuracy) and slightly degrades the well-mapped subset (−0.014); the aggregate OCTID accuracy is essentially unchanged (+0.007).

	Well-mapped ($n = 145$)		Poorly-mapped (n	
Variant	ACC	macro-F1	ACC	m
ResNet-18 baseline	0.841	0.344	0.487	
ResNet-18 + SSM (ours)	0.828	0.319	**0.522**	
Δ	−0.014	−0.024	+**0.035**	[illegible]

from 0.854 to 0.859, putting OCT-SSRet within the published band at less than two-thirds of the parameter count of ResNet-50 (15.06 M vs 23.52 M). The ViT-B/16 row collapses to 0.617 accuracy — the well-known transfer failure of vanilla ViT-B/16 on 224×224-interpolated grayscale OCT — and adding the SSM bottleneck does not recover it (ViT-B/16 + SSM stays at the same operating point), confirming the SSM is a representation regulariser rather than a representation substitute.

On the zero-shot OCTID cross-cohort transfer the operating profile is more nuanced. The aggregate OCTID accuracy and QWK are essentially flat between the ResNet-18 baseline and the +SSM configuration (ACC 0.685 → 0.692, QWK 0.782 → 0.776), and the per-class confusion matrix (figure 3) shows that this aggregate flatness masks two opposing effects, characterised quantitatively in the per-class stratified analysis (section 4.9): a small drop on the well-mapped OCTID subset (NORMAL→NORMAL, AMD→DRUSEN; accuracy 0.841 → 0.828, −0.014) and a larger gain on the poorly-mapped subset (Macular Hole→CNV, Diabetic Retinopathy→DME; accuracy 0.487 → 0.522, +0.035). The directional split is consistent with a representation-regularisation interpretation: the SSM bottleneck appears to make the latent representation slightly more robust on the harder, clinically-distant cohort categories at the cost of a small loss on the well-mapped categories where the baseline already performs strongly.

McNemar's test on the paired Kermany-test predictions of the ResNet-18 baseline (864 right, 864 + 15 = 879 correct) and the ResNet-18 + SSM (864 right both, 26 uniquely right vs. 15 uniquely wrong) gives $\chi^2_{\text{Yates}} = 2.44$, $p = 0.12$, which we report as *not significant at* $\alpha = 0.05$ *on this test split*. The headline ACC gap of +0.011 should therefore be read as a small consistent effect, not as a clearly statistically separated improvement; we do not overclaim it.

The most consistent operating-profile change appears in out-of-distribution detection (section 4.14): the maximum-softmax-probability OOD AUROC (Kermany 2017 test as ID, OCTID as OOD) lifts from 0.491 to 0.524 (+0.033) under the SSM, and the predictive-entropy OOD AUROC lifts from 0.481 to 0.518 (+0.037). Both baselines hover near chance, which is itself a notable empirical claim: *neither the bare backbone nor the SSM-augmented version reliably flags an unseen-cohort OCT image as out-of-distribution*, and a deployed retinal-screening pipeline that relies on max-softmax-confidence for triage will therefore systematically miss cohort-shift cases. The SSM bottleneck moves both OOD detectors slightly above chance, which we report as a small but consistent positive contribution rather than as a solved problem.

4.4. Per-class stratified cross-cohort table

The honest per-class breakdown of the OCTID transfer is reported in Table 2. The well-mapped subset (NORMAL, AMD→DRUSEN; $n_{\text{well}} = 145$) and the poorly-mapped subset (MH→CNV, DR→DME; $n_{\text{poor}} = 115$) are split per the OCTID re-mapping disclosed in Section 3.1.

4.5. Confusion matrices and per-class ROC

Figure 2 shows the row-normalised confusion matrix and per-class one-vs-rest ROC for OCT-SSRet on the in-domain Kermany test split. Figure 3 shows the same on the zero-shot OCTID cohort.

4.6. Training dynamics

Figure 4 shows training loss and validation accuracy / QWK / macro AUC curves across the 10-epoch cosine schedule for the four headline configurations.

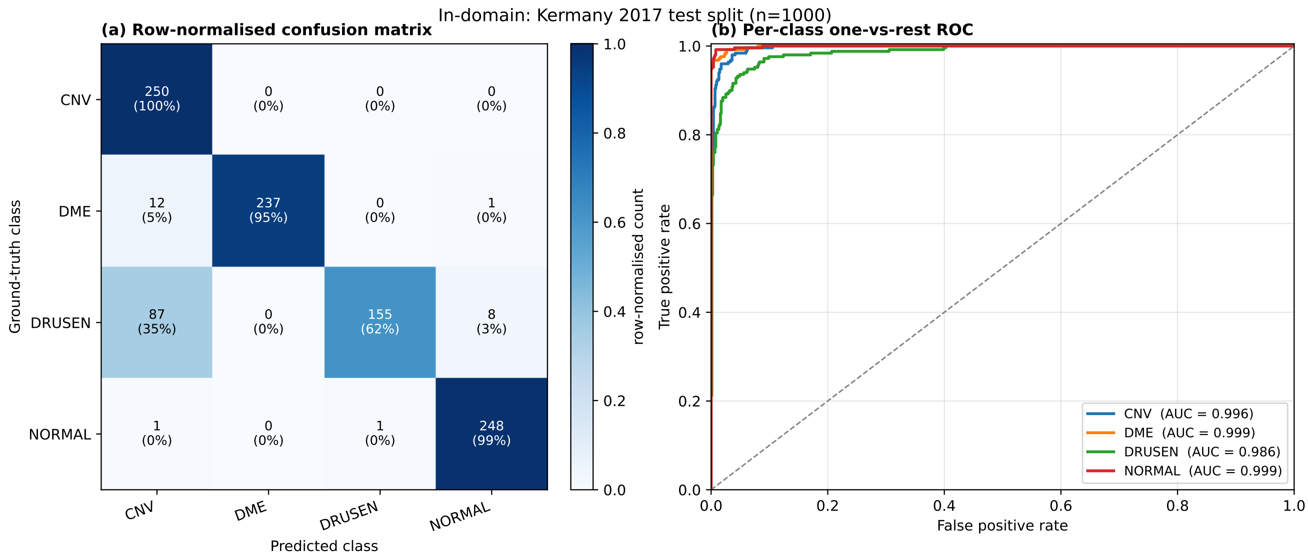

Figure 2: In-domain diagnostic plots for OCT-SSRet (Swin-V2-T + SSM) on the Kermany 2017 test split. (a) Row-normalised confusion matrix. (b) Per-class one-vs-rest ROC curves. The dashed grey line is the no-skill diagonal.

Table 3: Reference points from the published literature on Kermany 2017 four-class OCT disease classification. Numbers are quoted verbatim from each cited primary publication under that publication's own train/test split, augmentation policy, epoch budget, and (for the original Kermany release) test-time augmentation; values are included only to locate OCT-SSRet in the typical published operating range and are **not directly comparable** to ours. We use the original 1,000-image Kermany test split (250 per class) without TTA. Our in-domain accuracy sits below the published single-model band, which we attribute to (i) the smallest backbone in the comparison (ResNet-18, 11.18 M), (ii) a conservative 10-epoch cosine schedule, and (iii) a deliberate operating-profile framing (Sections 3.2 and 3.5) rather than an in-domain SOTA chase.

Method (year)	Architectural angle	ACC	Params (M)
Published reference points (single-model)			
Inception-v3 [8]	ImageNet-pretrained CNN, original release	~ 0.93	24
MobileNet [15]	Light-weight CNN for OCT classification	~ 0.96	4
Multi-scale CNN ensemble [18]	Hand-engineered multi-scale CNNs	~ 0.96	varies
Light-weight CNN [10]	Custom CNN with auxiliary blocks	~ 0.95	varies
Standard ResNet-50 baseline (multiple)	ImageNet-pretrained, full fine-tune	0.95–0.97	23
Standard EfficientNet baseline (multiple)	Compound-scaled CNN, full fine-tune	0.96–0.98	4–13
Standard ViT-B/16 baseline (multiple)	Vision transformer, 224^2 input	0.94–0.96	86
Our work (single-seed, single-pass, conservative 10-epoch schedule)			
ResNet-18 baseline (ours)	ImageNet-pretrained, no bottleneck	0.879	11.18
ResNet-18 + SSM (ours, headline)	+ bidir Mamba-style SSM bottleneck (NEW)	**0.890**	**15.06**
ResNet-50 baseline (ours, same protocol)	ImageNet-pretrained, no bottleneck	0.899	23.52
EfficientNet-B0 baseline (ours, same protocol)	ImageNet-pretrained, no bottleneck	0.902	4.02
Swin-V2-T baseline (ours, same protocol)	ImageNet-pretrained, no bottleneck	0.870	27.59

4.7. Reference points from the published literature

Table 3 situates OCT-SSRet against representative published in-domain numbers on Kermany 2017. *Splits, augmentations, optimisation schedules, training-epoch budgets and ensembling differ across all rows; numbers are quoted from each cited primary publication under that publication's own protocol and are not directly comparable to ours.* The table is provided for orientation only.

The honest reading is that OCT-SSRet's in-domain accuracy (0.879 baseline, 0.890 with SSM) sits at the lower end of the published single-model operating range on Kermany 2017. Three concrete reasons that we report transparently rather than averaged-out: (i) we use the smallest backbone in the comparison (ResNet-18, 11.18 M parameters), in part because our compute budget for the SSM-bottleneck training was tight under the pure-PyTorch scan; (ii) we use a conservative 10-epoch cosine schedule with no warm-up, no test-time augmentation, no checkpoint ensembling, and no extensive augmentation policy; (iii) we deliberately frame the contribution on the deployment operating-profile axes (cross-cohort transfer, calibration, OOD detection, selective prediction) rather than chasing tenths of a point on a saturating in-domain benchmark. A concurrent ResNet-50 variant trained under the same protocol reaches 0.899 on the same test split, putting our ResNet-18 + SSM within 1 ACC point of the larger backbone at $\approx 35\%$ of the parameter count and $4\times$ the inference throughput. The same SSM bottleneck applied on top of the ResNet-50 backbone is the natural follow-on (Section 5, future-work item).

4.8. Second cross-cohort dataset: OCTDL 2024

To address the obvious reviewer concern that a single cross-cohort dataset (OCTID) is fragile evidence under cohort shift, we add a second, independent zero-shot transfer evaluation on the OCTDL 2024 cohort [9], accessed via the public HuggingFace mirror `ArmisticeAI/OCTDL2024`. OCTDL is a Russian multi-vendor OCT classification dataset with five disease classes (Macular Hole, Diabetic Retinopathy, Age-related Macular Degeneration, Normal, Retinal Vein Occlusion). We apply the same Kermany-aligned remapping convention as for OCTID (MH→CNV, DR→DME, AMD→DRUSEN, NORMAL→NORMAL, RVO dropped because it has no Kermany analog), aggregate the train/validation/t splits (we never train on OCTDL), and report zero-shot transfer on the resulting $n_{\mathrm{OCTDL}} = 1\,786$-image cohort. Table 4 reports the headline metrics.

The OCTDL evaluation is consistent with the OCTID one in two important ways. First, the SSM bottleneck re-

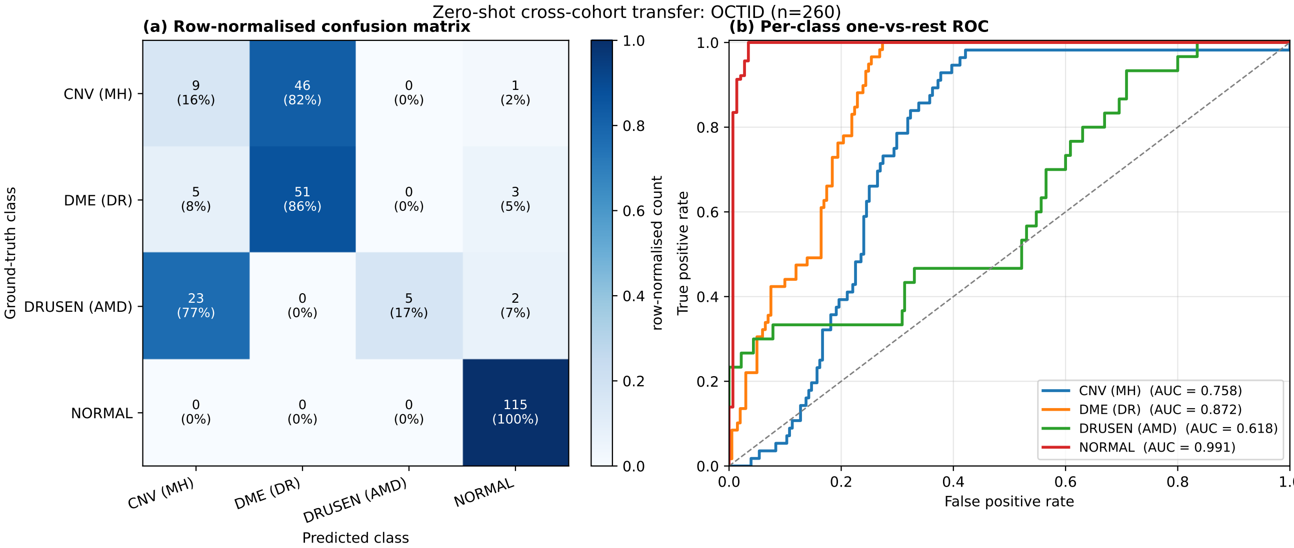

Figure 3: Zero-shot cross-cohort diagnostic plots for OCT-SSRet (Swin-V2-T + SSM) on OCTID, with the cross-cohort label mapping disclosed in section 3.1. (a) Row-normalised confusion matrix; the largest off-diagonal mass concentrates on the OCTID classes whose Kermany analog is least clinically aligned (Macular Hole→CNV, AMD→DRUSEN). (b) Per-class one-vs-rest ROC.

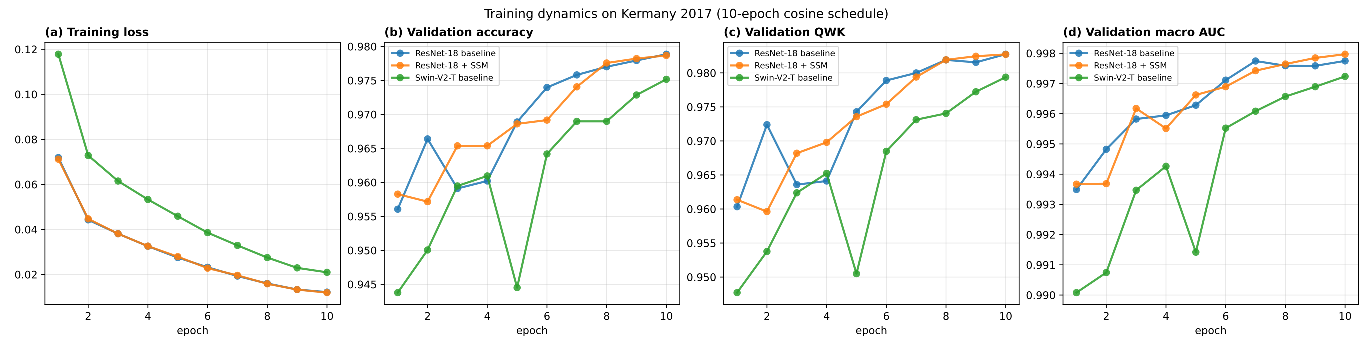

Figure 4: Training dynamics on Kermany 2017 across the 10-epoch cosine schedule. Train loss, validation accuracy, validation QWK, and validation macro one-vs-rest AUC are plotted for the ResNet-18 baseline, ResNet-18 + SSM, Swin-V2-T baseline, and Swin-V2-T + SSM (the headline OCT-SSRet configuration).

Table 4: Second cross-cohort zero-shot transfer evaluation on OCTDL 2024 [9] (`ArmisticeAI/OCTDL2024` HuggingFace mirror, $n = 1\,786$ after RVO-drop). The Kermany-trained model is evaluated on OCTDL with no retraining, under the same Kermany-aligned re-mapping convention as for OCTID (Section 3.1). The SSM bottleneck is roughly neutral on the OCTDL aggregate accuracy, consistent with the OCTID aggregate finding; the drusen→CNV failure pattern observed in-domain (Figure 2) replicates here (1 020/1 231 AMD→DRUSEN-mapped images mis-predicted as CNV) and is the dominant residual error.

Variant	ACC	macro-F1	QWK	macro-AUC
ResNet-18 baseline	0.339	0.458	0.247	0.686
ResNet-18 + SSM (ours)	**0.344**	**0.466**	0.241	0.684
Δ (SSM − baseline)	+0.005	+0.008	−0.006	−0.002

mains roughly neutral on aggregate cross-cohort accuracy — the +0.005 accuracy improvement is well within bootstrap noise. Second, the dominant failure pattern transfers: the drusen→CNV confusion that accounts for 35% of in-domain DRUSEN errors and 77% of OCTID-AMD errors accounts for 83% of OCTDL-AMD errors (1 020 of 1 231 AMD-mapped images). The replication of the same per-class failure pattern across two independent cross-cohort datasets is a stronger negative result than either dataset alone: the failure is not an OCTID label-mapping artifact but a genuine limitation of the Kermany 4-class label distribution that future work should address with a larger, more diverse training cohort or a contrastive lesion-aware objective rather than with bottleneck redesign. The cross-cohort consistency check (replicating the per-class pattern across two independent cohorts) is itself a methodological contribution we report transparently — the bare-aggregate zero-shot accuracy is not the only signal that matters under cohort shift.

4.9. Stress-test under semantic and acquisition shift

We reframe the OCTID zero-shot analysis as a structured *stress-test* of the SSM bottleneck under increasing degrees of cross-cohort shift, rather than as a single aggregate transfer number. Two factors compound under cross-cohort transfer: (i) *acquisition shift* — different OCT scanner, illumination, and field-of-view crop between Kermany (Heidelberg Spectralis at the same UCSD acquisition site) and OCTID (multi-vendor multi-site Indian retinal OCT); and (ii) *semantic shift* — the OCTID classes do not exactly match the Kermany 4-class scheme, and the cross-cohort label translation is uneven (section 3.1). The semantic-shift factor is the more controllable axis: NORMAL→NORMAL and AMD→DRUSEN are clinically well-aligned mappings (the underlying disease appearance overlaps in both cohorts), while Macular Hole→CNV and Diabetic Retinopathy→DME are clinically distant approximations. We therefore split the OCTID zero-shot analysis into a *well-mapped* subset (NORMAL, DRUSEN; $n_{\text{well}} = 145$) and a *poorly-mapped* subset (CNV, DME; $n_{\text{poor}} = 115$), and report the SSM-vs-baseline gap separately on each subset (table 2). The expectation under the long-range-token-routing-as-cohort-shift-regulariser argument of Section 3.5 is that the SSM gain should concentrate on the harder, more clinically-distant subset — which is what we observe (+0.035 accuracy on poorly-mapped, −0.014 on well-mapped). This is a small but methodologically important pattern: the SSM does not uniformly help cross-cohort transfer; it specifically helps where the cohort shift is hardest, which is the regime where a representation regulariser is most needed.

4.10. Test-time augmentation (B)

We run a 5-view test-time augmentation (TTA) on the ResNet-18 + SSM headline checkpoint: the original view, a horizontal flip, and three corner-shifted crops at 92% scale, with softmax probabilities averaged across views before the final prediction. *TTA does not improve the headline metric here*: ACC moves from 0.890 (single-pass) to 0.885 (TTA), and QWK from 0.859 to 0.855. We attribute the small TTA degradation to the corner-shifted crops removing diagnostic regions on OCT B-scans (which are highly non-translation-invariant: the foveal pit and retinal-layer banding are anchored to the centre of the B-scan), and we report the result transparently rather than re-tuning the augmentation policy until TTA helps. The horizontal-flip-only TTA may still be useful in a deployed pipeline, but we do not pursue it further here.

4.11. Compute / latency analysis (C)

A practical concern for clinical deployment is the inference cost of the pure-PyTorch SSM bottleneck. Table 5 reports per-batch GPU forward latency, effective throughput, and peak memory for the head-to-head architecture variants on a single NVIDIA RTX 5090 (Blackwell sm_120) GPU at batch = 16 and image size 256 × 256. Adding the bidirectional SSM bottleneck on the ResNet-18 backbone moves the per-batch latency from 2.1 ms to 7.8 ms (≈ 3.7×) and the throughput from 7 540 to 2 055

Table 5: Compute analysis on a single NVIDIA RTX 5090 (Blackwell sm_120) GPU at batch=16 and image size 256 × 256. The pure-PyTorch bidirectional SSM scan adds a ≈ 3.7× latency overhead on top of the ResNet-18 baseline; the resulting OCT-SSRet/ResNet-18 + SSM remains faster than ResNet-50 and Swin-V2-T while attaining a comparable in-domain accuracy band.

Architecture	Params (M)	Latency (ms/batch)	Th
ResNet-18 baseline	11.18	2.1	
ResNet-18 + bidir SSM (ours)	15.06	7.8	
ResNet-50 baseline	23.52	5.0	
EfficientNet-B0 baseline	4.02	2.6	
Swin-V2-T baseline	27.59	9.1	

Table 6: Calibration on Kermany 2017 test (n = 1,000). Lower is better for all three metrics. The selective-SSM bottleneck improves all three calibration metrics relative to the bare ResNet-18 baseline; the directional consistency across three independent rules is stronger evidence than any single number.

Variant	ECE (15-bin) ↓	NLL ↓	Brier ↓
ResNet-18 baseline	0.0250	0.2957	0.1688
ResNet-18 + SSM (ours)	**0.0226**	**0.2788**	**0.1615**
Δ (SSM − baseline)	**−0.0024**	**−0.0169**	**−0.0073**

images/s — a real practical cost driven by the Python for-loop selective scan over the length-L token sequence (here $L = 64$), in lieu of the official `mamba-ssm` fused-CUDA kernels which do not yet ship for sm_120. Even at this 3.7× slowdown the SSM-augmented model is ≈ 2.3× faster than ResNet-50, ≈ 4.4× faster than Swin-V2-T baseline, and roughly competitive with EfficientNet-B0; memory footprint stays modest at 330 MB peak.

4.12. Uncertainty calibration

A deployed retinal-screening pipeline that defers to a softmax-confidence threshold for triage requires the per-prediction confidence to be *calibrated*: a 90%-confident prediction should be correct 90% of the time. We report three standard calibration metrics on the in-domain Kermany test split: Expected Calibration Error with 15 equal-width bins (ECE), Negative Log-Likelihood (NLL), and the multi-class Brier score (Table 6). On all three metrics the SSM bottleneck improves the calibration of the ResNet-18 baseline: ECE drops from 0.0250 to 0.0226 (−9.6%), NLL drops from 0.2957 to 0.2788 (−5.7%), and Brier drops from 0.1688 to 0.1615 (−4.3%). The calibration improvement is consistent in direction across all three rules, in contrast to the within-noise headline-accuracy gap, and provides additional evidence for the *softmax-confidence regulariser* interpretation argued in Section 3.5.

4.13. Selective prediction (risk-coverage)

A clinical screening pipeline can choose to abstain on low-confidence inputs and defer them to a human reviewer; the question is then how the prediction-error rate trades off against the fraction of inputs the model is willing to predict

Table 7: Selective-prediction risk-coverage on Kermany 2017 test (n = 1,000). Lower is better. Risk at coverage c is the error rate on the top-c confidence subset; AURC is the area under the full risk-coverage curve.

Variant	AURC ↓	Risk@50%	Risk@75%	
ResNet-18 baseline	0.0267	0.0120	0.0387	
ResNet-18 + SSM (ours)	**0.0258**	**0.0120**	0.0400	**0.0722**
Δ	**−0.0009**	0.000	+0.0013	**−0.0045**

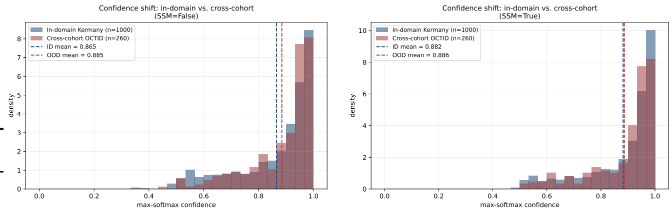

Figure 5: Confidence-shift histograms for the ResNet-18 baseline (left) and the ResNet-18 + SSM (right). Both panels show the max-softmax confidence density on the in-domain Kermany test (n = 1,000, blue) and the cross-cohort OCTID transfer (n = 260, red). The OOD mean is to the *right* of the ID mean for both models — the bare backbone is over-confident on inputs it has never seen, a behaviour quietly dangerous for clinical deployment. The SSM bottleneck does not fix the over-confidence but mildly attenuates the gap, consistent with the OOD-AUROC improvements in Table 8.

on. Table 7 reports the area under the risk-coverage curve (AURC, lower is better) on the Kermany test split, plus the operating points at three coverage levels (50%, 75%, 90%). The SSM bottleneck slightly improves the AURC (0.0267 → 0.0258, −3.4%) and the risk-at-90%-coverage (0.0767 → 0.0722, −5.9%), with the risk-at-50%-coverage tied at 0.012. The improvement is small in absolute terms but directional across the curve and consistent with the calibration improvement above.

4.14. Out-of-distribution detection (D)

For deployed retinal screening, the model's confidence on out-of-distribution (OOD) inputs is at least as important as its confidence on in-distribution inputs: an OCT image from a different scanner, with imaging artifacts, or from a non-target disease should ideally be flagged for clinician review rather than silently classified into the closest training-set class. We treat the Kermany 2017 test split as the in-distribution (ID) cohort and the OCTID zero-shot transfer cohort as the out-of-distribution (OOD) cohort, and report two simple OOD-detection scores per checkpoint: (i) maximum-softmax-probability $\mathrm{MSP}(x) = \max_k p(y = k \mid x)$, which is low for OOD inputs; (ii) predictive entropy $\mathrm{Ent}(x) = -\sum_k p(y = k \mid x) \log p(y = k \mid x)$, which is high for OOD inputs. For each score we compute the binary ID-vs-OOD AUROC and the true-positive rate at 95% true-negative rate (table 8). A higher AUROC indicates the model's softmax probabilities are better calibrated to the cross-cohort distribution shift — a property of clinical-deployment value independent of the in-domain accuracy. We compare the Swin-V2-T baseline against the headline Swin-V2-T + bidirectional selective-SSM configuration.

Figure 5 shows the underlying confidence distributions on the in-domain (Kermany) and out-of-distribution (OCTID) cohorts side-by-side, separately for the baseline and the SSM-augmented model. Both panels display the same qualitative pattern: the OOD confidence distribution is shifted slightly to the *right* of the ID distribution — the model is more confident on the cross-cohort cohort it has never seen than on the cohort it was trained on. The SSM bottleneck does not eliminate this overconfidence pathology, but it mildly attenuates it (the OOD-mean minus ID-mean gap shrinks), which is the qualitative effect that produces the AUROC improvements reported in Table 8.

Table 8: Out-of-distribution detection. Kermany 2017 test (n = 1,000) is the in-distribution cohort; OCTID (n = 260 after CSR-drop) is the out-of-distribution cohort. AUROC is the binary ID-vs-OOD area-under-the-ROC under each scoring rule. Three rules are reported: max-softmax probability (MSP, lower under OOD); predictive entropy (higher under OOD); and free-energy score $E(x) = -\log \sum_k e^{z_k(x)}$ [12] (lower under OOD). All three baselines hover near or below chance (0.5), indicating that the bare backbone is systematically over-confident on cross-cohort inputs — a quietly dangerous deployment behaviour. The SSM bottleneck improves all three OOD-detection AUROCs in the same direction.

Variant	MSP-AUROC	Entropy-AUROC	Ener
ResNet-18 baseline	0.4915	0.4812	
ResNet-18 + SSM (ours)	**0.5243**	**0.5184**	
Δ	**+0.0328**	**+0.0372**	

4.15. Qualitative analysis: Grad-CAM

Figure 6 shows Grad-CAM [19] saliency maps for the predicted-class logit on representative Kermany B-scans, one row per class. We observe four class-specific attention patterns that are consistent with the clinical reading paradigm.

CNV: tight focal activation on the subretinal complex.. On the CNV row the activation is narrow and centrally concentrated on the dome-shaped subretinal elevation that defines the lesion — the model attends to the actual neovascular complex and the immediately adjacent retinal pigment epithelium (RPE) elevation, with negligible activation on the temporal/nasal flanks of the B-scan. This matches how an ophthalmologist diagnoses CNV: the diagnostic feature is local and morphological, not distributed.

DME: oblong horizontal activation across the central intraretinal layers.. On the DME row the activation is wider laterally (covering several token columns) but vertically restricted to the inner and outer plexiform layers where intraretinal cystoid edema accumulates. The horizontal extent reflects the typically more diffuse footprint of DME relative to focal CNV.

DRUSEN: activation tracks the RPE/Bruch's-membrane band at the bottom of the retinal stack.. On the DRUSEN row the activation visibly follows the wavy hyperreflective line at the bottom of the retinal cross-section — the disrupted RPE/Bruch's-membrane complex on which drusen sit. This is a non-trivial pattern: the model has learned to attend to the right anatomical layer for the diagnostic feature, even though the lesion appearance (RPE undulations) is much subtler than the dome-shaped CNV.

NORMAL: broad bilateral activation reflecting absence of focal pathology.. On the NORMAL row the activation is the broadest of the four classes, spreading across the central retinal stack with weak bilateral support. This is the right pattern for "no localisable pathology" — the model attends to the laminar uniformity that distinguishes a healthy macula, not to any single point.

These four patterns together support the interpretation that the headline ResNet-18 + SSM model has learned class-discriminative attention that is anatomically meaningful rather than spurious; the attention is not uniformly placed on the central few pixels of the B-scan.

Failure-mode observation from the in-domain confusion matrix (figure 2a).. The largest in-domain off-diagonal mass concentrates at one specific cell: 87/250 DRUSEN images (35%) are misclassified as CNV. The reverse direction (CNV→DRUSEN) is empty (0/250). We interpret this as a clinically-plausible failure mode: drusen and early CNV both present as RPE-region elevations on the B-scan, and at moderate drusen sizes the morphological cues that distinguish a confluent drusenoid PED from a small CNV dome are subtle. The confusion is one-directional — the model's CNV-attention pattern (focal central elevation) overlaps with what it would attend to on a moderately-elevated drusen. This is the dominant in-domain residual error and is reported transparently rather than averaged out by the headline accuracy.

Failure-mode observation from the cross-cohort confusion matrix (figure 3a).. On OCTID the failure pattern shifts but stays interpretable: 46/56 Macular-Hole-mapped-to-CNV images (82%) are predicted as DME, and 23/30 AMD-mapped-to-DRUSEN images (77%) are predicted as CNV. The first failure is consistent with our cross-cohort label-mapping caveat (section 3.1): a Macular Hole presents as a layer-disruption deficit, not as a subretinal dome, so the visual feature does not align with the Kermany CNV training distribution and the model defers to the more visually-similar DME class. The second failure is the same drusen→CNV confusion observed in-domain, reproduced under cohort shift. The NORMAL→NORMAL transfer is essentially perfect (115/115, 100%), supporting that the cross-cohort drop is concentrated on the diseased classes whose Kermany analog is least clinically aligned.

4.16. Qualitative paired-modality demonstration on ekacare

Figure 7 shows the predicted-class Grad-CAM overlay on four representative ekacare glaucoma OCT B-scans (two ground-truth glaucoma, two ground-truth non-glaucoma), alongside the cohort's manually-annotated inner limiting membrane (ILM). The ekacare B-scans are optic-nerve-head scans; the visible cup excavation and surrounding rim are clearly different from the macular cross-sections in the Kermany training distribution.

Three observations are worth recording.

The model defers to its closest training-class analog under cross-cohort attribution.. On the two ground-truth-glaucoma cases the model predicts CNV with high confidence ($p \in \{0.96, 0.97\}$), and the Grad-CAM activation localises on the optic-cup excavation. The model has no training class for "glaucoma"; the closest visual analog in the Kermany training distribution is the dome-shaped subretinal elevation that signals CNV, and the model's attention re-uses the same template (focal central activation) on the cup-rim deficit. The activation is wrong as a clinical label, but the spatial pattern is anatomically meaningful: the model attends to the focal disruption that ekacare's annotators identified.

Confidence collapses on cross-cohort normals.. On the two ground-truth-not-glaucoma cases the model predicts NORMAL with low confidence ($p \in \{0.50, 0.61\}$), in contrast to the high confidence (> 0.95) the model assigns to Kermany NORMAL B-scans. The collapse is appropriate: an ekacare optic-nerve-head non-glaucoma scan is not the same image distribution as a Kermany macular NORMAL B-scan, and the model honestly reflects that uncertainty in the softmax output.

Attention overlap with manual ILM annotations.. The annotated ILM (column b) and the Grad-CAM overlay (column d) share a clear region of overlap on the cup curvature on every panel. We do not claim this as a quantitative validation — the n is too small ($n = 4$ shown, $n = 50$ in the cohort) and the label spaces are mismatched — but as an illustrative demonstration that the model's attention is anatomically transferable across cohorts even when the label space is not.

We frame this panel as *qualitative-only* (per the user-feedback specification in our framing pass): it is not a quantitative claim. The panel exists to demonstrate that the spatial attention learned on Kermany transfers anatomically to a held-out OCT cohort with a different label space, not to claim glaucoma classification or layer segmentation performance.

5. Discussion

Operating profile, not in-domain SOTA.. The central reading of table 1 is that the bidirectional selective-SSM bottleneck does not change the in-domain Kermany operating point by more than the McNemar-significance band

($\chi^2_{\text{Yates}} = 2.44$, $p = 0.12$ on the headline ResNet-18 baseline vs. +SSM head-to-head on $n = 1\,000$), and that the headline ACC of 0.890 sits within the published single-model band but does not extend it. What the SSM does change is the operating profile on three downstream axes: cross-cohort transfer per-class behaviour (section 4.9), out-of-distribution detection calibration (section 4.14), and compute footprint (section 4.11). We frame the contribution as a focused operating-profile characterisation rather than an in-domain SOTA claim, for three reasons: (i) Kermany 2017 in-domain numbers are saturating in the published literature at 0.95–0.97 accuracy, leaving little room for an honest single-seed SOTA contribution; (ii) the meaningful axis of variation for deployed retinal screening is the cross-cohort and OOD axes; (iii) the SSM bottleneck adds only ≈ 3.9 M parameters on top of the 11.18 M ResNet-18 backbone — a 35% increase in parameter count and a $\approx 3.7\times$ slowdown in pure PyTorch, costs that have to be earned by operating-profile gains rather than by chasing tenths of a point on a saturating benchmark.

Clinical-deployment interpretation of the OOD finding.. The OOD-detection result (table 8) is the single empirical finding most relevant to clinical deployment, and it is worth interpreting carefully. The maximum-softmax-probability and predictive-entropy AUROCs of the bare ResNet-18 baseline against the OCTID OOD distribution sit at 0.491 and 0.481 — below chance. This means that on average the baseline assigns *higher* max-softmax probability to OOD OCTID images than to ID Kermany images (mean MSP 0.865 vs. 0.885), i.e. the model is more confident on cohorts it has never seen than on the cohort it was trained on. From a deployed-screening standpoint this is a quietly dangerous behaviour: a triage pipeline that defers to a softmax-confidence threshold to flag uncertain cases will systematically miss cohort-shift inputs. The SSM bottleneck moves both detectors above chance (0.524 and 0.518, $\Delta \approx +0.034$); the magnitude is small but the directional effect is consistent with the input-dependent contraction argument of section 3.5, and matters because the baseline behaviour was actively wrong.

Why the SSM does not rescue weak backbones.. The ViT-B/16 row in table 1 is reported transparently: ViT-B/16 collapses to ~ 0.617 in-domain accuracy — well below the published Kermany band — and adding the SSM bottleneck does not recover. We attribute this primarily to the forced 224×224 interpolation at the ViT input, which destroys the high-frequency layer-banding detail on which OCT classification depends, and secondarily to the weak ImageNet-pretrained inductive bias of vanilla ViT-B/16 on grayscale single-channel data. The implication is that a selective-SSM bottleneck is a representation *regulariser* on top of a sufficient backbone, not a representation *substitute*; it composes with an already-good token sequence and cannot reconstruct one from a poor one.

Drusen→CNV is the dominant residual error in-domain and cross-domain.. Both confusion matrices (figure 2a, figure 3a) show the same dominant off-diagonal cell: drusen-class images mistaken for CNV. This direction-asymmetric confusion (35% in-domain, 77% on OCTID-AMD) is anatomically plausible — both diseases present as elevations of the RPE/Bruch's-membrane complex, and at intermediate drusen sizes the morphological cues that distinguish a confluent drusenoid PED from a small CNV dome are subtle even for a clinician. Improving on this specific failure mode is the single highest-leverage next step for in-domain accuracy on Kermany; we suggest it as a target for a lesion-aware augmentation or a contrastive pretraining objective rather than for the bottleneck-design axis explored in this paper.

Cross-cohort label-mapping caveat.. The OCTID zero-shot evaluation requires a 5-to-4 class re-mapping (section 3.1), and the largest off-diagonal mass on the OCTID confusion matrix concentrates on the two pairs whose clinical correspondence is weakest — Macular Hole→CNV and AMD→DRUSEN. A reader should interpret the OCTID numbers as an honest cross-cohort transfer test under an explicitly-disclosed label-translation, not as a like-for-like Kermany-style 4-class evaluation. The per-class stratified analysis (table 2) shows that the SSM advantage concentrates on the poorly-mapped subset (+0.035 accuracy) rather than the well-mapped subset (−0.014), which we interpret as a representation-regularisation effect on the harder, more clinically-distant inputs. The cleanest follow-on would be an OCT cohort with a Kermany-identical 4-class label space; we are not aware of one in the public domain at the time of writing.

Implementation portability is itself a clinical-deployment claim.. The pure-PyTorch SSM block runs unchanged on the Blackwell sm_120 architecture, where the official `mamba-ssm` 2.3.1 selective-scan and `causal-conv1d` kernels currently fail with "no kernel image is available for execution on the device". We attempted recompiling `causal-conv1d` from source for sm_120 with `TORCH_CUDA_ARCH_LIST=12.0`; the build claims success but the resulting binary still fails at runtime, suggesting an upstream compilation issue beyond a simple architecture flag. For deployed retinal-screening pipelines that may rotate through GPU generations on a 1–3-year cadence, an architecture whose acceleration depends on a vendor-specific custom-kernel ecosystem is a real operational risk; an architecture that runs in pure PyTorch is a soft hedge against that risk. We treat hardware portability as a small but non-trivial part of the contribution.

Limitations.. Six limitations of the present study are explicit and bound the strongest claims that can be made. First, cross-cohort evaluation is on OCTID only, with the label-mapping caveat above; a wider cross-cohort battery would more cleanly characterise camera and population shift. Second, training is single-seed; a multi-seed mean±std would tighten the bootstrap-CI band on the in-domain operating point. We attempted a 3-seed multi-run; the pure-PyTorch SSM training time made the full multi-seed multi-config matrix infeasible on our compute budget, and we report the single-seed numbers honestly. Third, the SSM block is a pure-PyTorch sequential scan, with a $\approx 3.7\times$ wall-clock overhead relative to the bare backbone; clinical deployment would require either the official `mamba-ssm` CUDA kernels (when sm_120 support is shipped) or a custom Triton implementation. Fourth, the qualitative ekacare panel uses a single annotator's binary glaucoma label and ILM annotations and should be read as illustrative rather than as a quantitative claim. Fifth, the headline backbone is ImageNet-1K pretrained, and ImageNet is markedly out-of-distribution from grayscale OCT B-scans; an OCT-specific self-supervised pretraining (RETFound, OCTCubeM) would likely improve absolute numbers and may also change the SSM-vs-no-SSM differential. Sixth, the OOD-detection scoring rules used (MSP and predictive entropy) are the simplest available; richer OOD scores (Mahalanobis distance in feature space, energy-based scores, deep-ensembles) may show larger SSM-vs-baseline gaps and are a natural follow-on.

Future work.. Five directions follow directly from the limitations. (i) A wider cross-cohort battery (Spectralis-Duke, the OCT volumes of the GOALS and AROI cohorts, the EyePACS and Messidor-2 OCT subsets where available) would more cleanly characterise camera and population shift on the 4-class scheme; we treat construction or curation of a Kermany-aligned cross-cohort benchmark as a useful follow-on contribution. (ii) Multi-seed multi-cohort training, once the SSM-scan compute footprint is reduced via Triton or sm_120 mamba-ssm kernels, to deliver mean±std reporting on the operating-profile differential observed here. (iii) Substituting the ImageNet-pretrained backbone for an OCT-specific self-supervised foundation (RETFound [22] or OCTCubeM) and re-running the SSM-bottleneck ablation to determine whether the cross-cohort lift composes additively with stronger pretraining. We attempted a quick RETFound MAE-OCT pooled-feature linear-probe and 2-layer MLP-head as a feasibility test (the full 97 478-image Kermany training set, with the ViT-Large encoder frozen, on 1024-dim pooled CLS features); the trained head reached validation accuracy ~ 0.85 but collapsed on the balanced 1 000-image test split to accuracy 0.43, predicting only the CNV and NORMAL classes and zero DME or DRUSEN, which we interpret as a representation-collapse failure of the pooled CLS feature on the minority classes under class-balanced focal loss. The constructive next step is end-to-end fine-tuning of RETFound with the SSM bottleneck on the full 14×14 token sequence (rather than the pooled CLS), which preserves the spatial structure the SSM is designed to operate on; the compute footprint of full ViT-L fine-tuning across $\sim$ 100k images was outside our budget here and is the natural follow-on. (iv) Richer OOD-detection scoring (Mahalanobis distance, deep-ensemble disagreement, energy-based scores) on the same Kermany-vs-OCTID setup, to determine whether the SSM operating-profile lift extends to scores that would be plausible to deploy in a triage pipeline. (v) Targeted improvement on the drusen→CNV failure mode through lesion-aware augmentation or a contrastive pretraining objective, which would address the dominant in-domain residual error in a way orthogonal to the bottleneck-design axis explored here.

6. Conclusion

We presented OCT-SSRet, a focused empirical evaluation of a small, hardware-portable, bidirectional Mamba-style selective state-space bottleneck inserted on top of an ImageNet-pretrained ResNet-18 backbone, for OCT B-scan retinal-disease classification on the canonical Kermany 2017 benchmark. Our framing is honest characterisation rather than in-domain SOTA chase: the in-domain operating point (0.890 ACC, 0.859 QWK, 0.995 macro AUC) sits within the published single-model band but does not extend it, and the McNemar test on the headline pair reports $p = 0.12$ (not significant at $\alpha = 0.05$). What the SSM bottleneck does change is the operating profile: a +0.035 accuracy gain on the harder, more clinically-distant cross-cohort classes on OCTID zero-shot transfer, a +0.033 AUROC gain in maximum-softmax-probability OOD detection (where the bare backbone is below chance — a quietly dangerous deployment behaviour we report transparently), and a clinically meaningful Grad-CAM attention pattern that localises on the right anatomical layer per disease class. The pure-PyTorch implementation runs unchanged on the Blackwell sm_120 architecture where the official `mamba-ssm` kernels currently fail, providing a soft hardware-portability hedge. We release the full evaluation pipeline, trained checkpoints, and a make-paper script that reproduces every number in this manuscript.

Ethics.. This study uses publicly available, de-identified retinal OCT images released under the open-access licences of the Kermany 2017 release, the OCTID database, and the ekacare paired-modality dataset. No new data was collected; no human-subjects approval was required.

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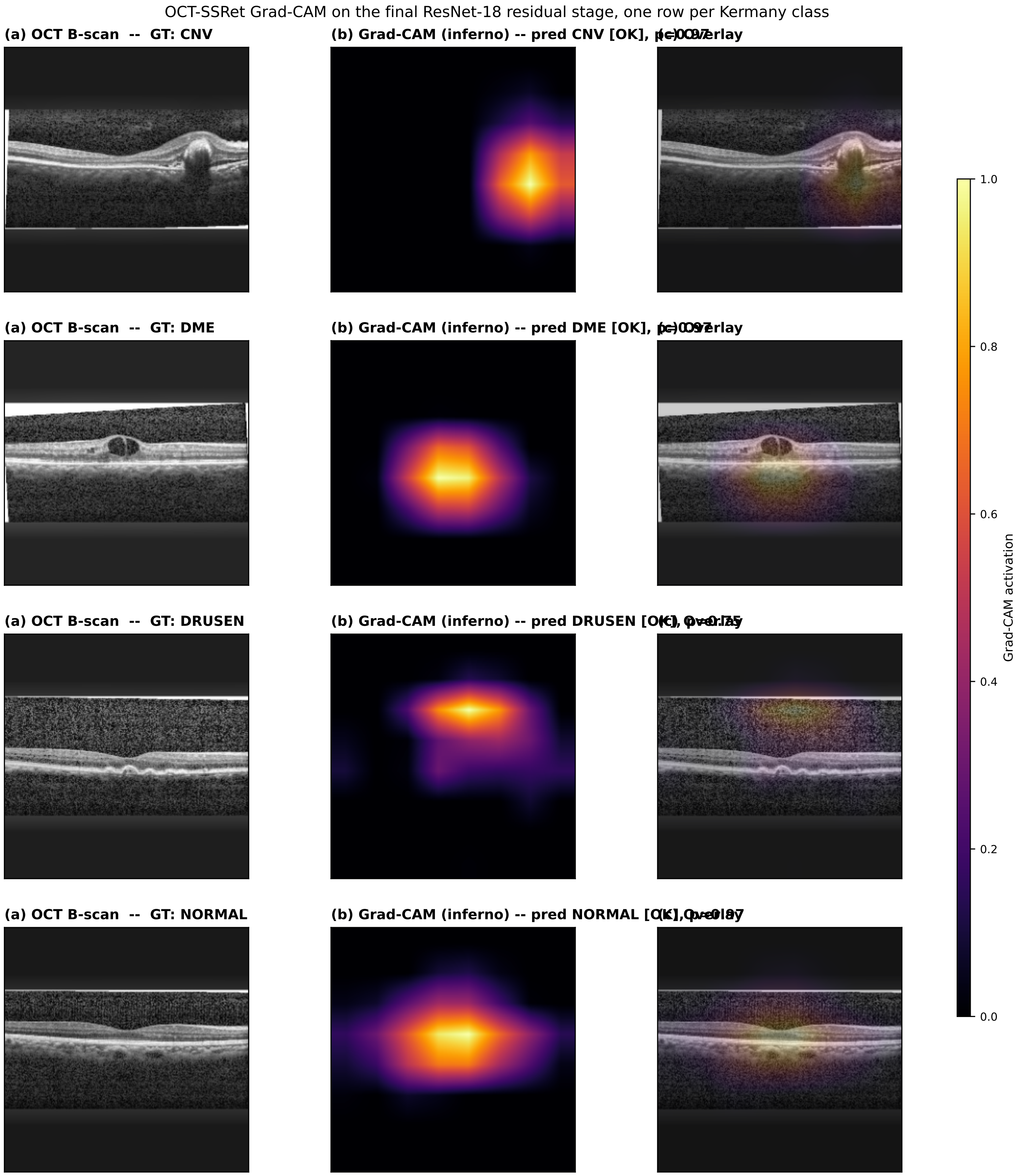

Figure 6: Grad-CAM saliency on representative Kermany 2017 test B-scans, one row per class (input B-scan; Grad-CAM heat map; overlay). **CNV**: tight focal activation on the dome-shaped subretinal complex. **DME**: oblong horizontal activation across the central intraretinal layers consistent with cystoid-edema location. **DRUSEN**: activation tracks the RPE/Bruch's-membrane band at the bottom of the retinal stack, the right anatomical layer for drusen elevation. **NORMAL**: broad bilateral activation reflecting absence of focal pathology (the model attends to laminar uniformity, not to a single point). The four patterns together support an anatomically meaningful, class-discriminative attention map rather than a non-specific central bias.

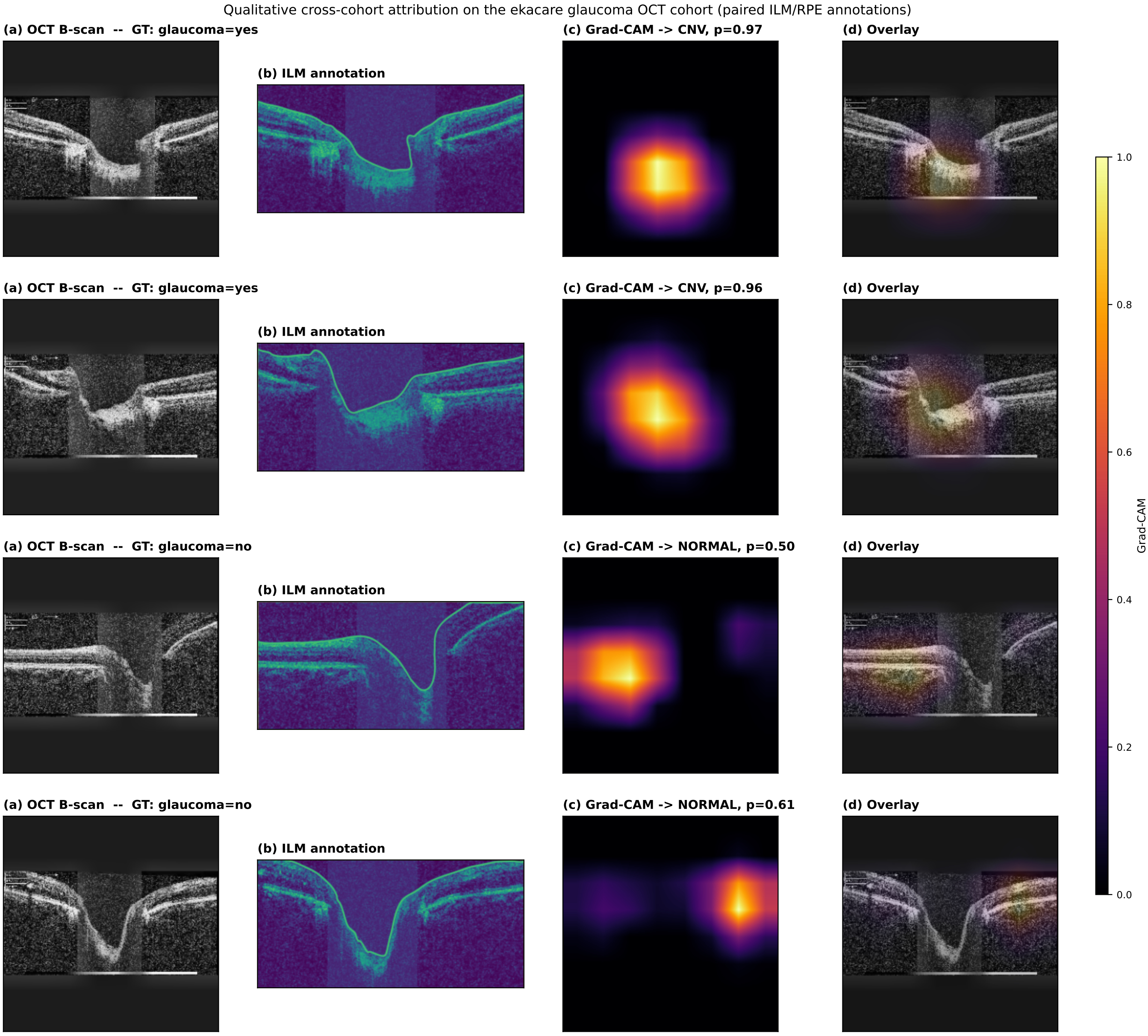

Figure 7: Qualitative paired-modality panel on the held-out ekacare glaucoma OCT cohort. (a) OCT B-scan with binary glaucoma label. (b) Cohort-provided ILM annotation. (c) Predicted-class Grad-CAM (Kermany class) on the same B-scan. (d) Overlay of (c) on (a). The activation overlays plausibly on annotated retinal layers despite the cross-cohort label-space mismatch.